

VAIBHAV MAHAPATRA

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EDUCATION

Johns Hopkins University

MS Robotics || CGPA: 4.0 / 4.0

May 2026

Baltimore, USA

Indian Institute of Technology, Madras

M.Tech Data Science + B.Tech Mechanical Engineering || CGPA: 8.75 / 10

May 2024

Chennai, India

- Secured an All India Rank of **3278 (99.73 percentile)** in JEE Advanced 2019 among **1.4M** candidates
- Selected among **top 10%** students offered opportunity to change Bachelors major, based on academic performance
- Granted a full-funded research fellowship of **USD 7500** as a 2023 MITACS Globalink Research Fellow

Technical Skills

- Coding languages:** Python, Julia, C++, MATLAB **Packages:** ROS2, Mujoco, PyTorch, OpenCV, YOLO, Git, Docker, TensorRT

PROFESSIONAL EXPERIENCE

AI Vision Engineering Co-Op

Asea Brown Boveri (ABB) Robotics

Jan 2026 – Present

San Jose, CA

- Productionizing **Random Bin Picking**, integrating CozyPose for accurate **6D pose estimation** to enhance robotic automation
- Built an LLM chat assistant with **GraphRAG** context injection, multi-session memory via vector embeddings, and TTFT of 3s
- Finetuned a π_0 text-conditioned **VLA** policy for real-time cloth folding with a dual 7-DOF arm controlled by **RWS** and **EGM**

Robotics Perception Intern

Scout Robotics (formerly, Sahay AI, Inc.)

Aug 2025 – Dec 2025

Philadelphia, PA

- Developed a **DeepLabV3+** backbone for semantic segmentation of the rail environment, achieving **83% mIoU** on test data
- Dockerized** and deployed a **TensorRT** engine on **AWS EC2 (GPU)**, delivering **2.19x** higher throughput for real-time inference
- Performed Lidar-Camera **sensor fusion**, enabling accurate monocular depth estimation with RGB and precise PointClouds

Software Engineering Intern

The MathWorks, Inc.

May 2025 – Aug 2025

Boston, MA

- Accelerated Stateflow system testing by converting Simulink models to C++, achieving a **99%** reduction in test latency
- Automated Simulink-to-C++ code generation using a **custom LLM**-powered pipeline, reducing development effort by **4 weeks**
- Secured podium finishes in **3** internal hackathons, including *Vision Bird*, a computer vision-driven real-time MATLAB game

Language Modeling Engineer

Securities and Exchange Board of India (SEBI)

Nov 2021 – Jan 2022

Mumbai, India

- Predicted market manipulation by scraping text messages from open Telegram channels using their public API
- Engineered a **BERT** based **Transfer Learning** Model for word classification with an **F1** score of **0.82**
- Assembled a **CamemBERT** word embedding to execute regression for sentence segmentation on text messages

Machine Learning Intern

Digital Outcomes Technologies Pvt. Ltd.

May 2021 – Jul 2021

Mumbai, India

- Built **Collaborative filtering** based Recommender engine to increase revenue by **USD 4000** for a B2B eCommerce client
- Utilized sklearn to build multiple KNN models benchmarked on RMSE - deployed the recommender with **RESTful API**

RESEARCH EXPERIENCE

Self-supervised reversal of Motion Blur with NAFNet | Masters Thesis

Prof. A. N. Rajagopalan, Department of Electrical Engineering, IIT Madras

Aug 2023 – June 2024

Chennai, India

- Employed a **NAFNet** CNN architecture to de-blur motion-blurred footage of static scenes from a Go-Pro dataset
- Assembled a **self-supervised** architecture, extracted depth and imposed photometric losses to secure a **PSNR** of **31.2**
- Presented a 1 hr Seminar on **Diffusion** Models, based on a literature survey on **25+** research papers, to a lab of 40 scholars

Scientific ML with ACE to estimate inter-atomic potentials

Prof. Christoph Ortner, Institute of Applied Mathematics, University of British Columbia

May 2023 – Aug 2023

Vancouver, Canada

- Modeled inter-atomic potentials of *Si* and *TiO₂* in Julia utilizing the **Atomic Cluster Expansion (ACE)** approach
- Engineered a **Hierarchical Multilevel** learning approach to regularise the Bayesian Ridge Regression fit

Turbulent Super-Resolution GAN for Fluid flow modeling

Prof. Heinz Pitsch, Institute for Combustion Technology, RWTH Aachen University

May 2022 – Jul 2022

Aachen, Germany

- Re-engineered a **Super-Resolution GAN (ESRGAN)** for turbulence modeling, enhancing simulated resolution & fidelity
- Utilized **PINNs** to impose constraints and achieve a **92% accuracy** in predicting functional relations from filtered DNS data

TECHNICAL PROJECTS

- UR5 Robot Camera Calibration and Path Planning | Prof. Simon Leonard** *Sept 2024*
- Performed **hand-eye calibration** in MATLAB with Park and Martin Method for a gripper-camera setup
 - Implemented **RRT-connect** in C++ for a UR5 robotic arm, as a replacement for the MoveIt planner for motion planning
 - Simultaneously Localized and Mapped a TurtleBot's environment with an Extended Kalman Filter (**EKF**) and Particle Filter
- Robotic Arm Manipulation with Online Reinforcement Learning | Prof. Enrique Mallada** *Oct 2024*
- Developed and trained a UR5e robotic arm in the **Mujoco** simulation environment using OpenAI's **Stable-Baselines3**
 - Designed state and action spaces and implemented **reward shaping** to optimize training for multiple manipulator agents
 - Benchmarked the real-time performance of agents trained with **four RL algorithms**: SAC, A2C, TD3, and PPO
- Low-Light Mosquito Recognition | Prof. Kaushik Mitra** *Apr 2024*
- **Won 3rd place** in the hackathon among 150+ participants across over 45 teams
 - Modeled atmospheric haze and executed a SOTA **de-hazing** algorithm to enhance low-light mosquito images
 - Trained a novel ensemble of **YOLOv8** for detection with Inception V3 for classification to obtain a **mAP** score of **61.3%**
- NYC Parking Violation Estimation | Cloud Computing** *Apr 2023*
- Modeled NYC parking ticket violation in **Google Cloud** platform harnessing big data of **22M** instances and 43 attributes
 - Trained a model with **PySpark** and streamed real-time data with **Kafka**; attained an **F1** score of **0.92** with an accuracy of 96%
- Credit Fraud Detection Hackathon | American Express** *Oct 2022*
- **Won** the hackathon with **2nd place** among **120+ participants** across 50+ teams by modeling the credit history of users
 - Benchmarked **RandomForest**, **XGBoost**, **LightGBM** and **CatBoost** classifiers to estimate default rate and achieve an **F1** of **0.67**
- Automated Cardiac Segmentation and Ejection Fraction Estimation | Prof. Jerry L. Prince** *Apr 2025*
- Trained a **UNet** for automated left ventricular (LV) segmentation from echocardiogram videos of over 7,460 patient recordings
 - Engineered an empirical area-to-volume relationship to accurately estimate ejection fraction (EF), achieving an RMSE of **1.42%**
 - Secured the **best** EF estimation performance among **40+** students through rigorous evaluation, validating the robust pipeline
- Diffusion Tensor Neuroimage Registration and Synthesis | Prof. Jerry L. Prince** *Apr 2025*
- Benchmarked SimpleITK, ANTSpy, and FireANTS pipelines for GPU parallelization in **SyN** Diffeomorphic registration
 - Implemented 3D UNet and CNN architectures like **AttentionUNet** and **SegResNet** for voxel-to-voxel synthesis of DTI MRIs
- Contact-Aware Control of 2-Link Hopper via Residual Dynamics Learning | Prof. Joseph L. Moore** *May 2025*
- Developed a hybrid control system for a 2-link hopper using NN **dynamics residuals** to model complex contact forces
 - Used **Proximal Policy Optimization** to generate trajectory data and train residuals, improving simulation accuracy and balance
 - Integrated the hybrid dynamics model with a Model Predictive Path Integral (**MPPI**) controller for real-time control of the robot

LEADERSHIP EXPERIENCE

- Teaching Assistant - Algorithms for Sensor-Based Robotics & Robot Kinematics, Dynamics and Control** *2025*
- Devised problem sets, hosted office hours, conducted midterms, and computed participation statistics for **240+** students
- Corporate Relations Executive | Shastra Technical Fest, IIT Madras** *May 2020 – Jun 2022*
- Raised INR 800K (USD 10K) to strategically execute 3 hackathons, 2 quizzes, a PR campaign and a fireside chat